SE8394

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MARIAN COLLEGE KUTTIKKANAM (AUTONOMOUS)

UG Programme (Honors) Degree October 2025 Regular Examinations

Third Semester

243CCRUBC204 Design and Analysis of Algorithms

Time: **1.5 Hours** Maximum Marks: **30**

Answer one question from each bunch

BUNCH I

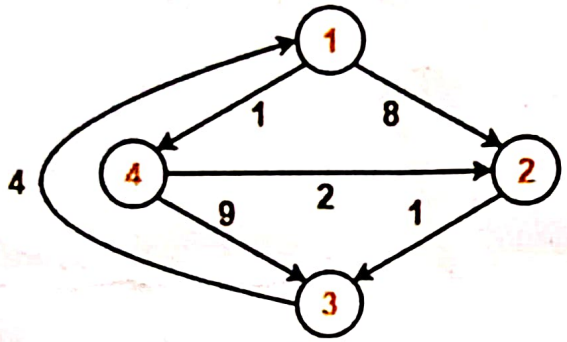
I.A A company wants to monitor the memory usage of its algorithms. Explain space complexity and how it influences algorithm design. (CO1:10 Marks)

OR

I.B A software company is developing a mobile application that must handle user requests efficiently as the number of users grows. The team has two candidate algorithms for a key operation: one with time complexity O(n) and another with $O(n^2)$. As the lead developer, you are asked to recommend which algorithm should be chosen.
Explain your decision using asymptotic notations. IAlso clearly describe the best, average, and worst-case scenarios of both algorithms and justify the choice using Θ (Theta) notation. Provide reasoning on how this affects performance for large inputs. (CO1:10 Marks)

BUNCH II

II.A



A city traffic authority has mapped the travel times (in minutes) between four important intersections (1, 2, 3, and 4). The routes are one-way, and the travel times are shown on the edges of the graph. The authority wants to know the shortest travel time between every pair of intersections so that drivers can be provided with the most efficient routes. Using the given graph, apply the Floyd–Warshall algorithm step by step to compute the shortest paths between all pairs of intersections and present the final shortest-path matrix.

(CO2:10 Marks)

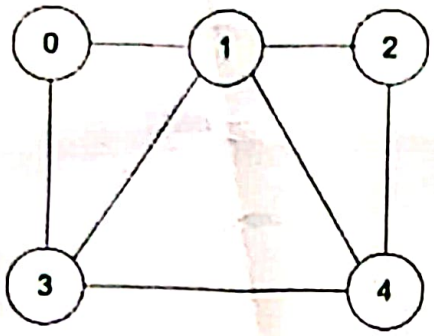
OR

II.B A metropolitan transport authority is studying the travel times between major bus terminals in the city. The routes are one-way, and the travel times are represented as edge weights in a directed graph. Since passengers may need to transfer through intermediate terminals, the authority wants to know the shortest travel times between every pair of terminals.How can the Floyd–Warshall algorithm be applied in this scenario to find the all-pairs shortest paths? Present the algorithm and explain its working with reference to the given situation

(CO2:10 Marks)

BUNCH III

III.A



A city tourism department wants to design a sightseeing bus route that starts at the main terminal, visits every tourist spot exactly once, and finally returns to the terminal without repeating any spot. The network of tourist spots and roads is represented by the given graph.

As a computer science student, explain how the concepts of a Hamiltonian Path and a Hamiltonian Cycle apply to this problem. Using the backtracking method, show how to check whether such a cycle exists in the given graph and provide one valid Hamiltonian Cycle if possible.

(CO3:10 Marks)

OR

III.B During a university chess fest, the organizing team needs to arrange queens on an N×N chessboard so that none of them can attack each other. Initially, they think of testing every possible placement, but soon realize it is highly inefficient as the number of possibilities grows rapidly. In this context, explain how the backtracking approach provides an effective solution to the N-Queens problem and why it is considered more efficient than brute-force search.

(CO3:10 Marks)